

Lab 02 Template

1.) Screenshot of cd/ls practice

(10 points)

Ans)

```
Ubuntu 22.04 LTS ns1 tty1
ns1 login: cpre230
Password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sun Feb  9 08:36:00 PM UTC 2025

System load:  0.0               Processes:            200
Usage of /:   47.3% of 9.75GB   Users logged in:     0
Memory usage: 25%              IPv4 address for ens160: 42.49.30.200
Swap usage:   0%

0 updates can be applied immediately.

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection
or proxy settings

Last login: Sun Feb  9 20:29:58 UTC 2025 on tty1
cpre230@ns1:~$ pwd
/home/cpre230
cpre230@ns1:~$ cd /
cpre230@ns1:/$ ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  sys  usr
boot  etc  lib  lib64  lost+found  net  proc  run  snap  swap.img  tmp  var
cpre230@ns1:/$ cd var
cpre230@ns1:/var$ ls var
ls: cannot access 'var': No such file or directory
cpre230@ns1:/var$ ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp
cpre230@ns1:/var$
```

2.) Screenshot of directory structure (10 points)

```
System load: 0.0          Processes:           202
Usage of /:  47.3% of 9.75GB Users logged in:       0
Memory usage: 25%        IPv4 address for ens160: 42.49.30.200
Swap usage:  0%

0 updates can be applied immediately.

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection
or proxy settings

Last login: Sun Feb  9 20:36:00 UTC 2025 on tty1
cpre230@ns1:~$ pwd
/home/cpre230
cpre230@ns1:~$ mkdir Documents
cpre230@ns1:~$ mkdir Pictures
cpre230@ns1:~$ mkdir Music
cpre230@ns1:~$ mkdir bin
cpre230@ns1:~$ ls
bin  Documents  Music  Pictures
cpre230@ns1:~$ cd Pictures
cpre230@ns1:~/Pictures$ mkdir pets
cpre230@ns1:~/Pictures$ mkdir report
cpre230@ns1:~/Pictures$ mkdir catz
cpre230@ns1:~/Pictures$ cd ..
cpre230@ns1:~$ mv Pictures/report Documents/
cpre230@ns1:~$ cd Documents
cpre230@ns1:~/Documents$ ls
report
cpre230@ns1:~/Documents$ cd ..
cpre230@ns1:~$ mv Pictures/catz Pictures/kittens
cpre230@ns1:~$ cd Pictures
cpre230@ns1:~/Pictures$ ls
kittens  pets
cpre230@ns1:~/Pictures$ _
```

3.) Difference between > and >> (10 points)

```
cpre230@ns1:~$ cat Documents/sample.txt > Documents/new_file.txt
cpre230@ns1:~$ cat Documents/new_file.txt
"Hi this is Pulkit and I am taking cpre 2300. I am a sophomore studying at the Iowa State."

cpre230@ns1:~$ cat Documents/sample.txt >> Documents/new_file.txt
cpre230@ns1:~$ cat Documents/new_file.txt
"Hi this is Pulkit and I am taking cpre 2300. I am a sophomore studying at the Iowa State."

"Hi this is Pulkit and I am taking cpre 2300. I am a sophomore studying at the Iowa State and my maj
or is computer science."

cpre230@ns1:~$
```

>: It copies the content that is written in the sample.txt to new_file.txt.

>>: It adds some new content to new_file.txt without deleting the previous data.

4.) How do you search in less? (10 points)

Ans) We can search in less by typing '/' followed by the word we want to search for(forward searching) and '?' followed by the word you are searching for(backward searching). We can move forward by just typing 'n' and we can move backwards by typing 'N'.

5.) What do which, tail, head, and grep do?

(10 points)

Ans) tail: It prints the specified number of lines from the end of a file.

head: It prints the specified number of lines from the top of the file.

grep: It searches amongst all the lines in a file and prints the lines that follow that pattern.

Which: It returns pathnames of files which have to be executed in the current environment, provided its arguments are given as commands in the shell.

6.) Screenshot of /var/log/auth.log

(10 points)



```
cp230@ns1:~$ cd /var/log
cp230@ns1:/var/log$ tail -n 10 /var/log/auth.log
Feb 10 04:29:19 ns1 sudo: cp230 : TTY=ttty1 ; PWD=/home/cpre230 ; USER=root ; COMMAND=/usr/bin/nano Documents/sample.txt
Feb 10 04:29:19 ns1 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cp230(uid=1000)
Feb 10 04:29:26 ns1 sudo: pam_unix(sudo:session): session closed for user root
Feb 10 04:32:12 ns1 sudo: cp230 : TTY=ttty1 ; PWD=/home/cpre230 ; USER=root ; COMMAND=/usr/bin/nano Documents/sample.txt
Feb 10 04:32:12 ns1 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cp230(uid=1000)
Feb 10 04:33:19 ns1 sudo: pam_unix(sudo:session): session closed for user root
Feb 10 04:51:58 ns1 sudo: pam_unix(sudo:auth): authentication failure; logname=cp230 uid=1000 euid=0 tty=/dev/tty1 ruser=cp230 rhost= user=cp230
Feb 10 04:52:08 ns1 sudo: cp230 : TTY=ttty1 ; PWD=/home/cpre230 ; USER=root ; COMMAND=/usr/bin/cat /etc/shadow
Feb 10 04:52:08 ns1 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cp230(uid=1000)
Feb 10 04:52:08 ns1 sudo: pam_unix(sudo:session): session closed for user root
cp230@ns1:/var/log$ _
```

7.) How does one view all processes using ps?

(10 points)

Ans) We can type the command 'ps aux' or 'ps -aux' to view all processes using ps.

8.) Screenshot of top output, name of process with PID 1
(10 points)

```
top - 05:43:16 up 3 days, 9:58, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 198 total, 1 running, 197 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.3 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 971.2 total, 62.6 free, 200.4 used, 708.2 buff/cache
MiB Swap: 1888.0 total, 1887.7 free, 0.3 used. 614.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5353	root	20	0	0	0	0	I	0.3	0.0	0:00.03	kworker/u2:1-events_power_e+
5357	cpres230	20	0	10504	3908	3308	R	0.3	0.4	0:00.24	top
1	root	20	0	102056	12980	8212	S	0.0	1.3	0:06.28	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.04	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
9	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
10	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_tasks_rude_
11	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_tasks_trace
12	root	20	0	0	0	0	S	0.0	0.0	0:00.33	ksoftirqd/0
13	root	20	0	0	0	0	I	0.0	0.0	0:07.61	rcu_sched
14	root	rt	0	0	0	0	S	0.0	0.0	0:00.47	migration/0
15	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
18	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
19	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	inet_frag_wq
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
21	root	20	0	0	0	0	S	0.0	0.0	0:00.60	khungtaskd
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
23	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	writeback
24	root	20	0	0	0	0	S	0.0	0.0	0:05.86	kcompactd0
25	root	25	5	0	0	0	S	0.0	0.0	0:00.00	ksmd
26	root	39	19	0	0	0	S	0.0	0.0	0:00.00	khugepaged
72	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kintegrityd
73	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kblockd
74	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	blkcg_punt_bio
75	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	tpm_dev_wq
76	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	ata_sff

Systemd is the process/command that is used with PID1.

9.) Method of running ping in bg without output on screen
(20 points)

Ans) We can use 'ping XXX.XXX.XXX.254 -i 5 > /dev/null 2>&1 &' for running ping in background without displaying its output on the screen.